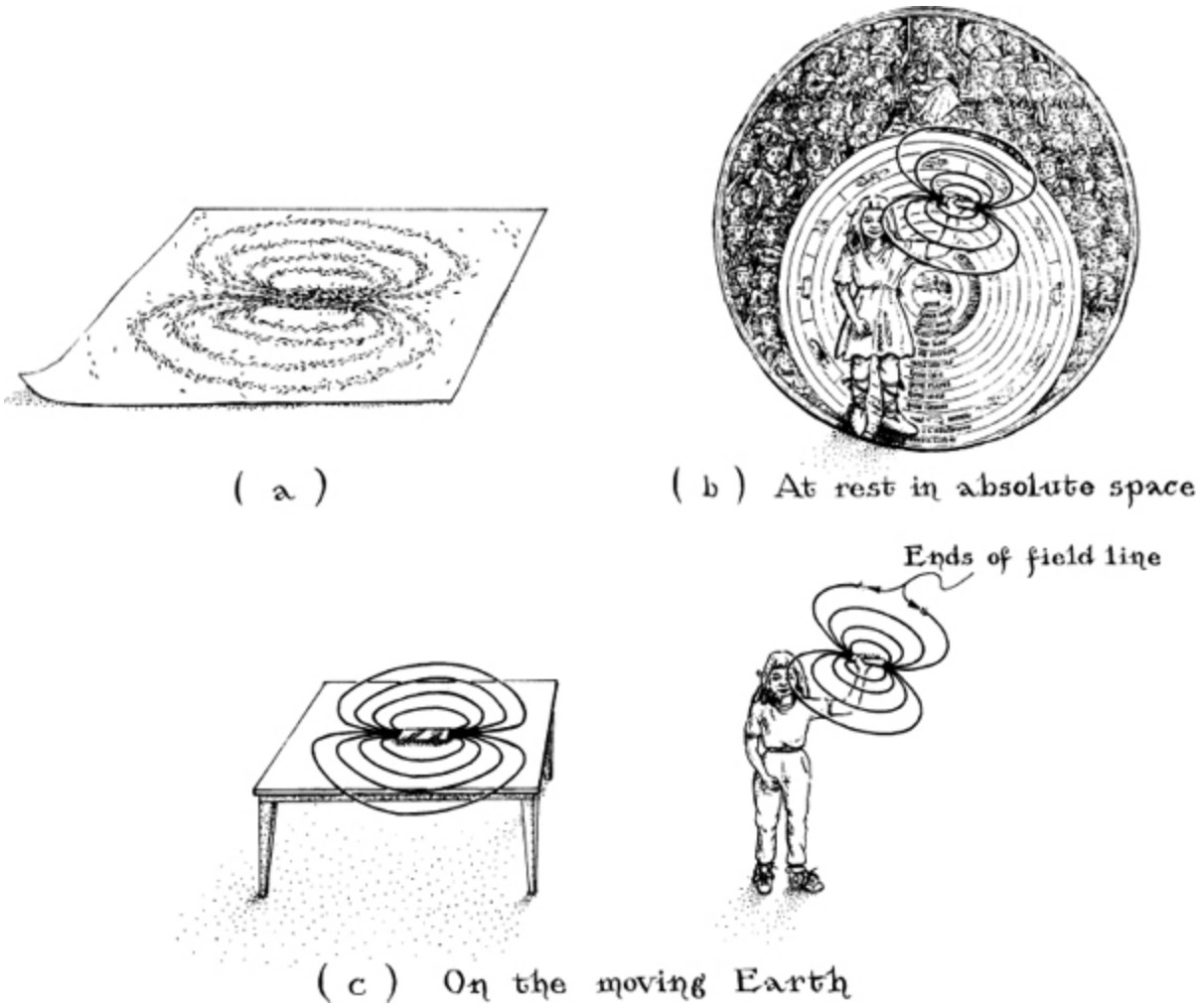
time flows more slowly as measured by someone moving through the Universe than by someone at rest; motion “dilates” time.



1.1 One of Maxwell’s electromagnetic laws, as understood within the framework of nineteenth-century, Newtonian physics: (a) The concept of a magnetic field line: When one places a bar magnet under a sheet of paper and scatters iron filings on top of the sheet, the filings mark out the magnet’s field lines. Each field line leaves the magnet’s north pole, swings around the magnet and reenters it at the south pole, and then travels through the magnet to the north pole, where it attaches onto itself. The field line is therefore a closed curve, somewhat like a rubber band, without any ends. The statement that “magnetic field lines never have ends” is Maxwell’s law in its simplest, most beautiful form. (b) According to Newtonian physics, this version of Maxwell’s law is correct no matter what one does with the magnet (for example, even if one shakes it wildly) *so long as one is at rest in absolute space.* No magnetic field line *ever* has any ends, from the viewpoint of someone at rest. (c) When studied by someone riding on the surface of the Earth as it moves through absolute space, Maxwell’s law is much more complicated, according to Newtonian physics. If the moving person’s magnet sits quietly on a table, then a few of its field lines (about one in a hundred million) will have ends. If the person shakes the magnet wildly, additional field lines (one in a trillion) will get cut temporarily by the shaking, and then will heal, then get cut, then reheal. Although one field line in a hundred million or a